BIG DATA ANALYTICS / MINI PROJECT

AgriPulse

From farm data to smarter decisions

A data-driven agriculture intelligence concept for crop, climate and soil insights.

PROJECT PRESENTATION • 2026

Presented by: Navaneethan | Department: AI & Data Science

A farm produces data. But does it produce decisions?

Farm decisions are often made with scattered information: rainfall reports, soil readings, crop history and market signals.

SOIL

Moisture and nutrient readings may be collected at different times.

WEATHER

Rainfall and temperature can change quickly across locations.

CROP

Crop choice and growth depend on multiple connected factors.

DECISIONS

Without combined analysis, useful patterns can be missed.

CORE PROBLEM

How can we combine agricultural datasets and turn them into clear, timely insights?

Meet AgriPulse: one view of the farm

AgriPulse is a proposed analytics pipeline that brings soil, weather and crop data together to support evidence-based farm planning.

COLLECT

Public datasets
+ sample sensor data

→

CLEAN

Handle missing values,
units and duplicates

→

ANALYZE

Find patterns,
compare conditions

→

PRESENT

Dashboards +
practical insights

OUTPUT **A simple decision-support dashboard — not an automated replacement for farmer expertise.**

Three data streams. One agricultural picture.

01 WEATHER

Rainfall • temperature • humidity

Use: seasonal patterns and rainfall variability.

02 SOIL

Soil moisture • pH • N-P-K nutrients

Use: compare soil conditions across fields.

03 CROP

Crop type • yield • sowing/harvest dates

Use: compare crop performance over time.

POSSIBLE DATA SOURCES

NASA POWER: weather and agroclimatology observations.

FAOSTAT: agriculture production and yield statistics.

Soil data: government/open soil datasets or clearly labelled sample sensor readings.

For the prototype, start with downloadable CSV files. Add live sensor feeds only if hardware is available.

How the data moves through the system

DATA SOURCES — Weather CSV | Soil CSV | Crop/yield CSV

↓

INGESTION — Python loader → optional Kafka stream

↓

STORAGE — Raw files + cleaned Parquet / CSV

↓

PROCESSING — Apache Spark / PySpark transformations

↓

ANALYTICS — Aggregations • correlations • trend analysis

↓

USER OUTPUT — Streamlit dashboard / Power BI report

Big-data element: distributed processing with Spark; Kafka is an optional extension for streaming.

What will the project actually calculate?

RAINFALL PATTERNS

Monthly totals, seasonal variation and unusually dry periods.

SOIL CONDITION

Average moisture, pH distribution and nutrient comparisons.

CROP PERFORMANCE

Yield by crop, location and season; compare trends.

RELATIONSHIP ANALYSIS

Explore correlations between rainfall, soil readings and yield.

LOCATION COMPARISON

Compare districts/regions using the same time period.

RISK FLAGS

Highlight low rainfall or soil readings outside chosen ranges.

Important: correlation can reveal relationships, but it does not prove that one factor caused a yield change.

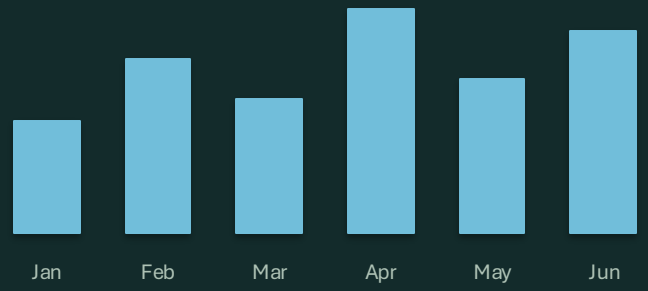
A dashboard designed around farmer questions

AGRIPULSE / FIELD OVERVIEW

Illustrative dashboard mock-up • values below are sample data, not live measurements

RAIN IN PERIOD	AVG SOIL MOISTURE	CROP IN VIEW	YIELD RECORDS
142 mm	28%	Paddy	1,240

MONTHLY RAINFALL (SAMPLE)



INSIGHT CARDS

01 Rainfall is highest in April in this sample.

02 Compare soil moisture before scheduling irrigation.

03 Filter by district, crop and season to explore patterns.

A realistic mini-project implementation plan

1 → 2 → 3 → 4

WEEK 1

Collect 2–3 datasets; inspect columns, formats and coverage.

WEEK 2

Clean data; standardize dates, units, location names and missing values.

WEEK 3

Run PySpark analysis; create summary tables and trend calculations.

WEEK 4

Build dashboard; validate results; prepare demo and documentation.

MINIMUM VIABLE DEMO

Upload CSV → process with PySpark → filter by crop/region → show 3 insights.

How will we know it works?

DATA QUALITY

Missing-value rate
Duplicate-row count
Valid date/location coverage

PROCESSING

Time taken on a fixed dataset
Compare single-machine pandas vs Spark (where meaningful)

USEFULNESS

Can a user filter a crop/region?
Can the dashboard explain the pattern clearly?

RESPONSIBLE USE

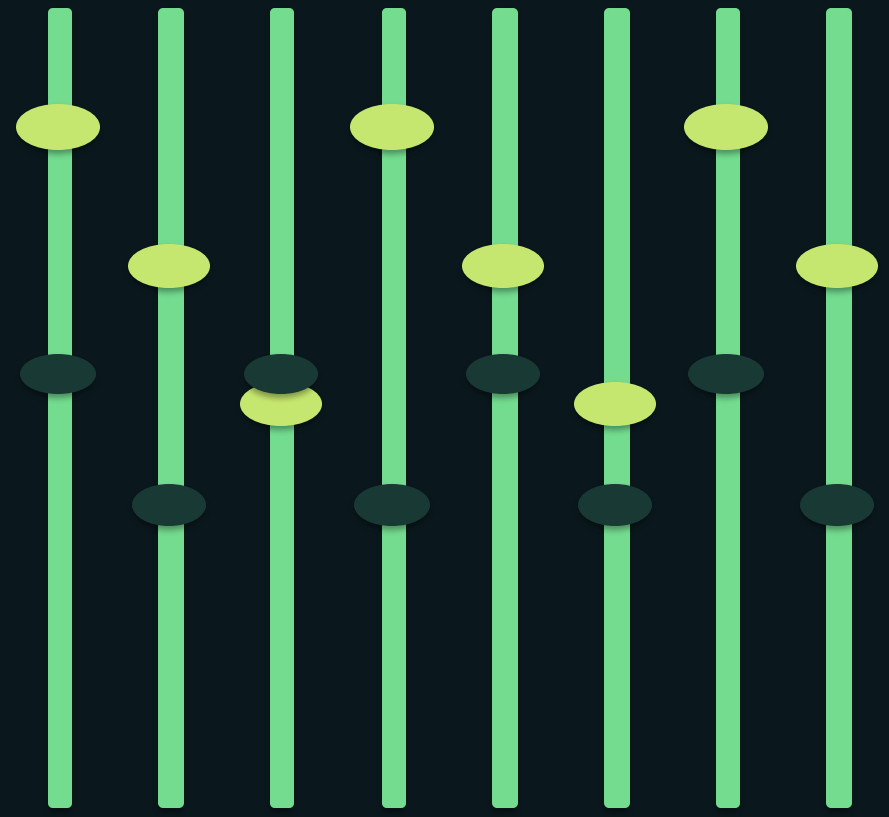
Use public or consented data. Label sample values clearly. Protect farm/location details. Treat insights as decision support, not guaranteed crop or yield advice.

Scope note: this is an analytics prototype; recommendations must be validated with local agronomy expertise.

THE TAKEAWAY

Better farm decisions begin with better data.

AgriPulse connects weather, soil and crop records into one analytics workflow — making patterns easier to see and decisions easier to discuss.



THANK YOU

Questions?